



**SIMATS SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES CHENNAI-602105**



Web-Based Student Skill Assessment and Development Management System

A Assignment Problem

Submitted in the partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING IN

Computer Science and Engineering

Submitted by

**Enamala Gowtham (192311190)
A. Ganesh Ram Kamal (192311183)
G. Abhinay (192373062)**

Under the Supervision of

Dr. Sudhagar. D

COURSE ASSIGNMENT

A. Assignment Information

Particular	Details
Department	CSE
Programme	B.Tech
Course Code & Course Name	CSA4305 – INTERNET PROGRAMMING
Academic Year / Batch	2026-27
Faculty Name	Dr.D.Sudhagar
Assignment Title	Student Skill Development Management System
Date of Issue	01.09.2026
Date of Submission	03.09.2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Develop visually appealing web pages using CSS for layout and style, and create dynamic, interactive functionality with JavaScript. CO2: Build dynamic web applications using XML, XSLT, and JSP within the MVC paradigm for efficient data handling and presentation.
Bloom's Taxonomy Level	L4 – Analyze / L5 – Evaluate / L6 – Create
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Supports systematic tracking of student skills, certifications, projects, training and employability readiness for institutions and industry-oriented education.

1. Problem Statement and Problem Formulation

Problem Statement

Students develop skills through academic courses, projects, internships, certifications, workshops and other learning activities. However, this information is often maintained in different places, making it difficult to obtain a unified view of a student's skill development. It is also difficult to identify skill gaps, recommend suitable learning activities and measure improvement over time.

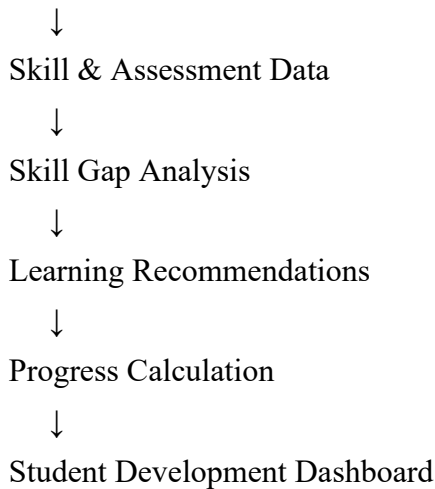
The proposed Student Skill Development Management System provides a centralized web-based platform for students, faculty and mentors to manage and monitor student skill development. The system supports skill-profile creation, skill categorization, assessment, certification and project tracking, learning recommendations and progress monitoring.

Problem Formulation

The system accepts student details, skills, proficiency levels, assessment scores, certifications and project information as input. It processes this information to identify competency gaps and calculate progress.

Input → Processing → Output

Student Data



The major stakeholders are students, faculty/mentors and administrators.

2. Objectives and Expected Outcomes

Objectives

The main objectives of the project are:

1. To develop a centralized student skill-development platform.
2. To create and maintain student skill profiles.
3. To categorize student skills.
4. To assess student proficiency levels.
5. To identify gaps between current and required skill levels.
6. To recommend suitable learning activities.
7. To track certifications, projects and internships.
8. To monitor student progress through a dashboard.
9. To provide a simple and user-friendly web interface.
10. To apply Internet Programming concepts such as HTML, CSS, JavaScript, XML and XSLT.

These objectives correspond with the assignment's requirement to support skill profiles, assessments, evidence, certifications, projects, recommendations and progress monitoring.

Expected Outcomes

The expected outcomes are:

- A functional web-based student skill management system.
- Centralized student skill information.
- Automatic skill-gap identification.
- Assessment result classification.
- Certification and project tracking.
- Learning recommendations.
- Progress monitoring.
- Easy access to student development information.

3. Requirements, Constraints and Assumptions

3.1 Functional Requirements

The system provides the following functions:

S.No	Requirement
1	Student registration
2	Student login
3	Profile management
4	Skill management
5	Skill categorization
6	Skill assessment
7	Skill-gap identification
8	Learning recommendations
9	Certification tracking
10	Project/internship tracking
11	Progress monitoring
12	Dashboard generation

The assignment also identifies learning-resource tracking, mentor feedback, goal setting, notifications and report generation as broader functional requirements.

3.2 Non-Functional Requirements

Performance

The system should provide responsive interaction and timely dashboard updates.

Usability

The application should have simple navigation and easy-to-use forms.

Reliability

The system should prevent duplicate or inconsistent skill records.

Security

Student information should be protected through authentication, authorization and input validation.

Scalability

The system should allow additional students, skills and learning resources to be added.

3.3 Constraints

- Limited academic development time.
- Limited development resources.
- The application should use approved web technologies.
- Student information should be protected.
- The system should be modular and maintainable.
- The prototype should be easy to execute in a standard web browser.

These constraints are consistent with the assignment's technical, security, reliability, cost/time and scalability requirements.

3.4 Assumptions

1. Every student has a unique registration number.
2. Skill levels are represented on a scale of 1–5.
3. Assessment scores range from 0–100.
4. Required skill levels are defined for each skill.
5. Students enter valid information.
6. Skill recommendations are based on identified gaps.
7. The prototype stores data using browser Local Storage.

4. Application of Relevant Course Knowledge / Concepts

The project applies several Internet Programming concepts.

HTML5

HTML is used to create the structure of:

- Login page
- Registration page
- Dashboard
- Assessment page
- Certification page
- Project page
- Forms
- Tables
- Navigation menus

CSS3

CSS is used for:

- Page layout
- Colors and typography
- Buttons
- Cards
- Tables
- Forms
- Responsive design

JavaScript

JavaScript is used to implement:

- Form validation
- Login validation
- Skill addition
- Duplicate-skill checking

- Assessment processing
- Skill-gap calculation
- Progress calculation
- Certification management
- Project management
- Dashboard updates
- Local Storage operations

XML

XML is used to store structured skill information.

XSLT

XSLT transforms XML skill data into an HTML-based presentation.

MVC Concept

The system can be extended using MVC:

Model → Student, Skill, Assessment, Project, Certification

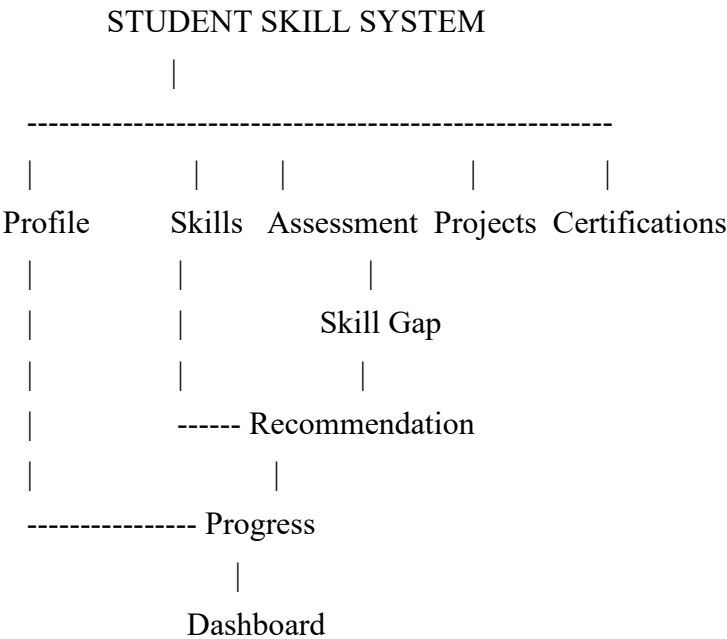
View → HTML + CSS pages

Controller → JavaScript / future JSP-Servlet controller

The assignment requires application of web/application development, data-management and system-design principles and recommends entities such as Student, Skill, Assessment, Certification, Project, Learning Activity, Mentor, Feedback and Progress History.

5. Design / Proposed Solution / Methodology

5.1 System Architecture



5.2 Modules

Module 1 – Login and Registration

Allows students to create accounts and securely access the application.

Module 2 – Student Profile

Stores:

- Name
- Register number
- Email
- Department
- Year
- Career goal

Module 3 – Skill Management

Students can add skills and specify current and required proficiency levels.

Module 4 – Assessment

Assessment scores are converted into proficiency levels.

Module 5 – Skill Gap Analysis

The system compares current proficiency with the required proficiency.

Module 6 – Recommendations

Skills with gaps are identified and learning activities are recommended.

Module 7 – Certifications

Certification name, organization, date and certificate ID are maintained.

Module 8 – Projects

Academic projects, personal projects, internships and workshops can be recorded.

Module 9 – Dashboard

Displays skill statistics, skill gaps and overall progress.

6. Algorithm / Pseudocode / Flowchart

6.1 Skill Gap Algorithm

Algorithm

1. Start.
2. Read current skill level.
3. Read required skill level.
4. Calculate the skill gap.
5. If the gap is greater than zero, mark the skill as requiring improvement.
6. Otherwise mark the skill as achieved.
7. Display the result.
8. Stop.

Pseudocode

START

INPUT currentLevel

INPUT requiredLevel

gap = requiredLevel - currentLevel

IF gap > 0 THEN

 status = "Improvement Required"

ELSE

 status = "Requirement Achieved"

END IF

DISPLAY gap

DISPLAY status

STOP

6.2 Assessment Algorithm

START

INPUT skill

INPUT score

IF score >= 80

 level = "Advanced"

ELSE IF score >= 60

 level = "Intermediate"

ELSE IF score >= 40

 level = "Beginner"

ELSE

 level = "Needs Improvement"

END IF

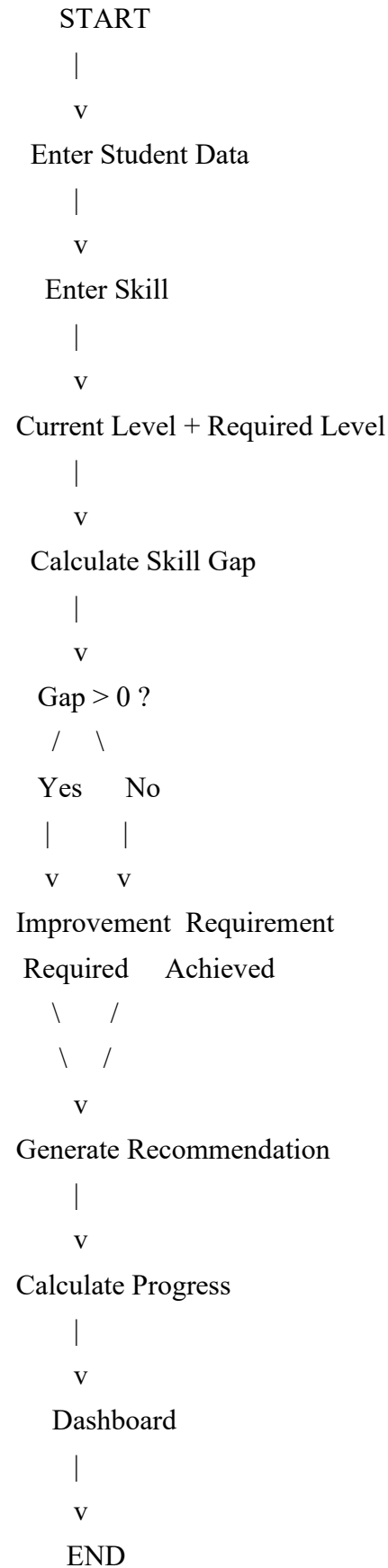
DISPLAY skill

DISPLAY score

DISPLAY level

STOP

6.3 Flowchart



7. Implementation / Source Code and Environment / Tools Used

7.1 Software and Tools

Tool/Technology	Purpose
HTML5	Page structure
CSS3	Styling and responsive interface
JavaScript	Dynamic functionality
XML	Structured skill data
XSLT	XML transformation
LocalStorage	Prototype data storage
Google Chrome	Testing
Microsoft Edge	Testing
VS Code/Notepad	Code development
Git/GitHub	Version control

The assignment recommends technologies such as HTML/CSS/JavaScript, REST APIs, database systems, UML/ER modelling and Git/GitHub where appropriate.

7.2 Project Structure

Program

```

|
├── index.html
├── login.html
├── registration.html
├── dashboard.html
├── assessment.html
├── certification.html
├── projects.html
├── style.css
├── script.js
├── skills.xml
├── skills.xsl
└── README.md

```

7.3 Implementation

The implementation consists of separate HTML pages for each major module. CSS provides a common interface, while JavaScript connects the modules and manages data.

Browser LocalStorage is used in the prototype to retain information between pages.

8. Test Cases and Expected / Actual Results

Test ID	Test Case	Input	Expected Result	Actual Result
TC01	Valid Login	Correct credentials	Login successful	Pass

Test ID	Test Case	Input	Expected Result	Actual Result
TC02	Invalid Login	Wrong password	Error displayed	Pass
TC03	Registration	Valid details	Account created	Pass
TC04	Add Skill	Java, level 4/5	Skill added	Pass
TC05	Duplicate Skill	Existing Java	Warning displayed	Pass
TC06	Skill Gap	3 and 5	Gap = 2	Pass
TC07	Assessment	Score 85	Advanced	Pass
TC08	Assessment	Score 65	Intermediate	Pass
TC09	Certification	Valid details	Certification added	Pass
TC10	Project	Valid project	Project added	Pass
TC11	Dashboard	Multiple skills	Statistics displayed	Pass
TC12	Progress	Skill records	Progress calculated	Pass

9. Execution Screenshots / Outputs

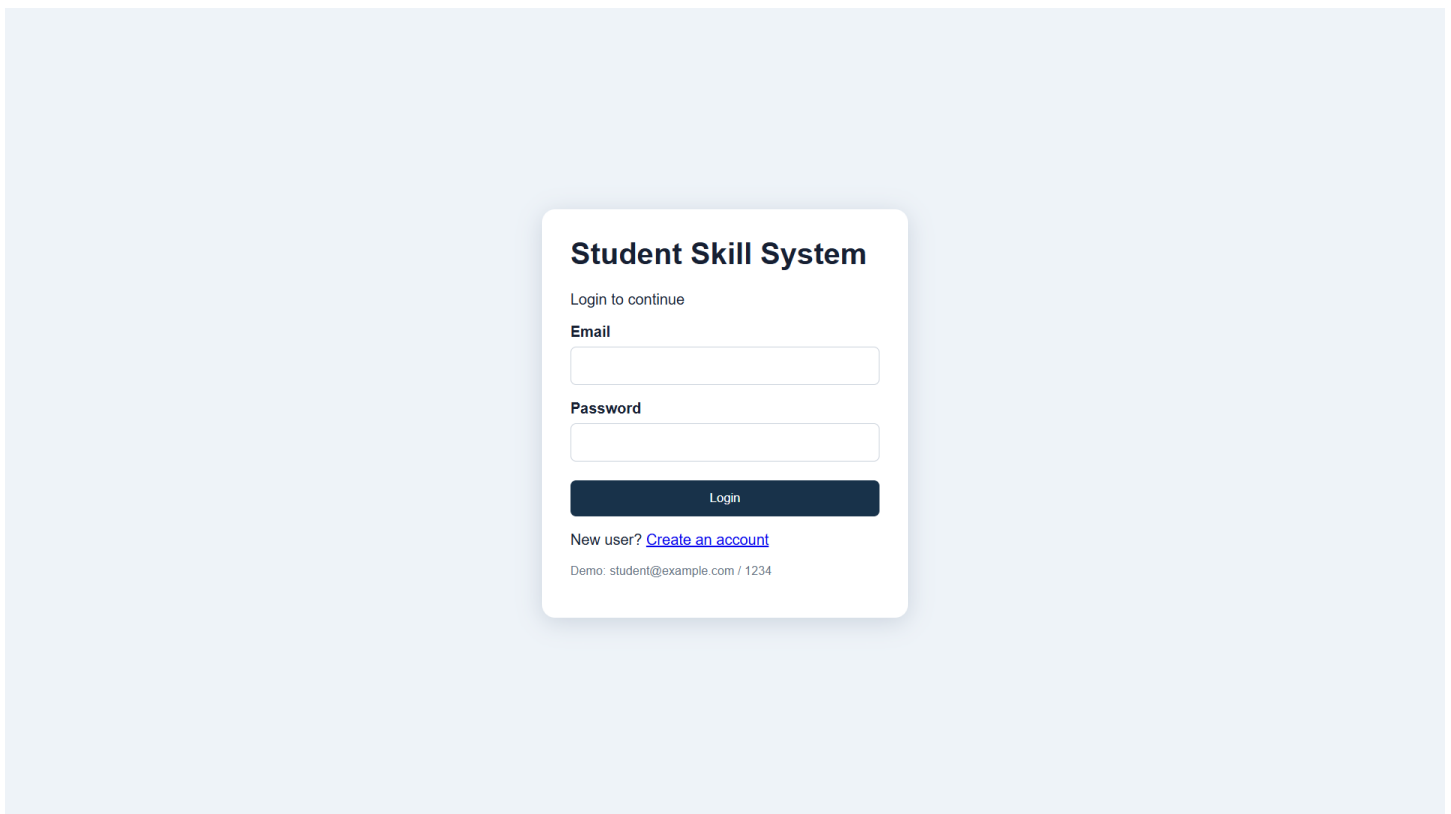
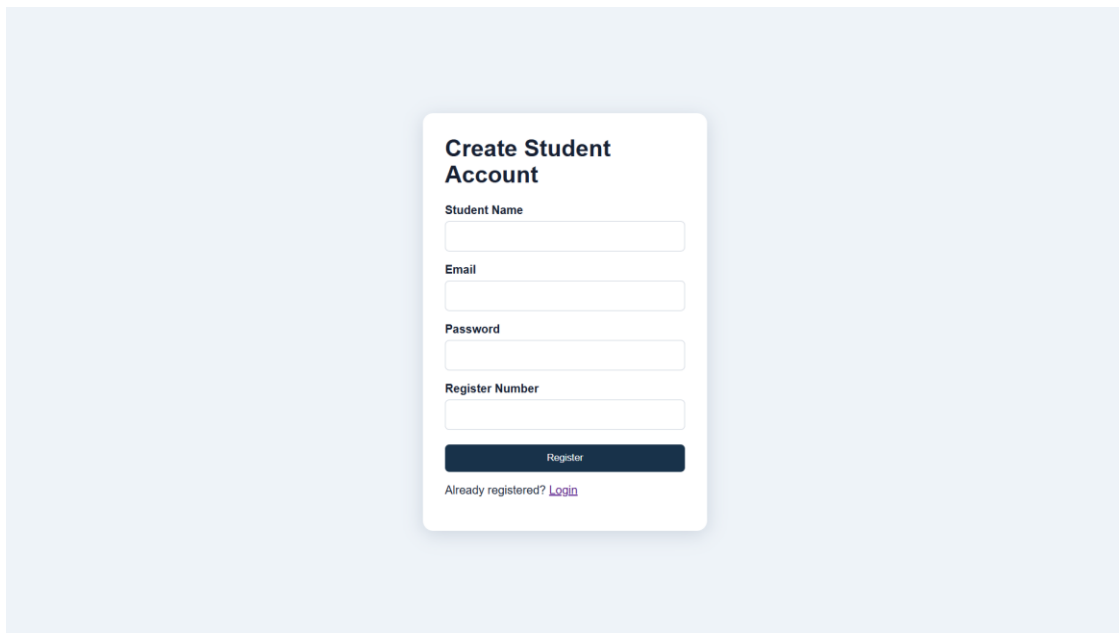


Figure 1 – Login Page

Output: Login page displays email and password fields with a login button.



The image shows a 'Create Student Account' form centered on a light blue background. The form is a white card with a dark blue header. It contains five input fields: 'Student Name', 'Email', 'Password', and 'Register Number'. Below these fields is a dark blue 'Register' button. At the bottom of the card, there is a link that says 'Already registered? Login'.

Create Student Account

Student Name

Email

Password

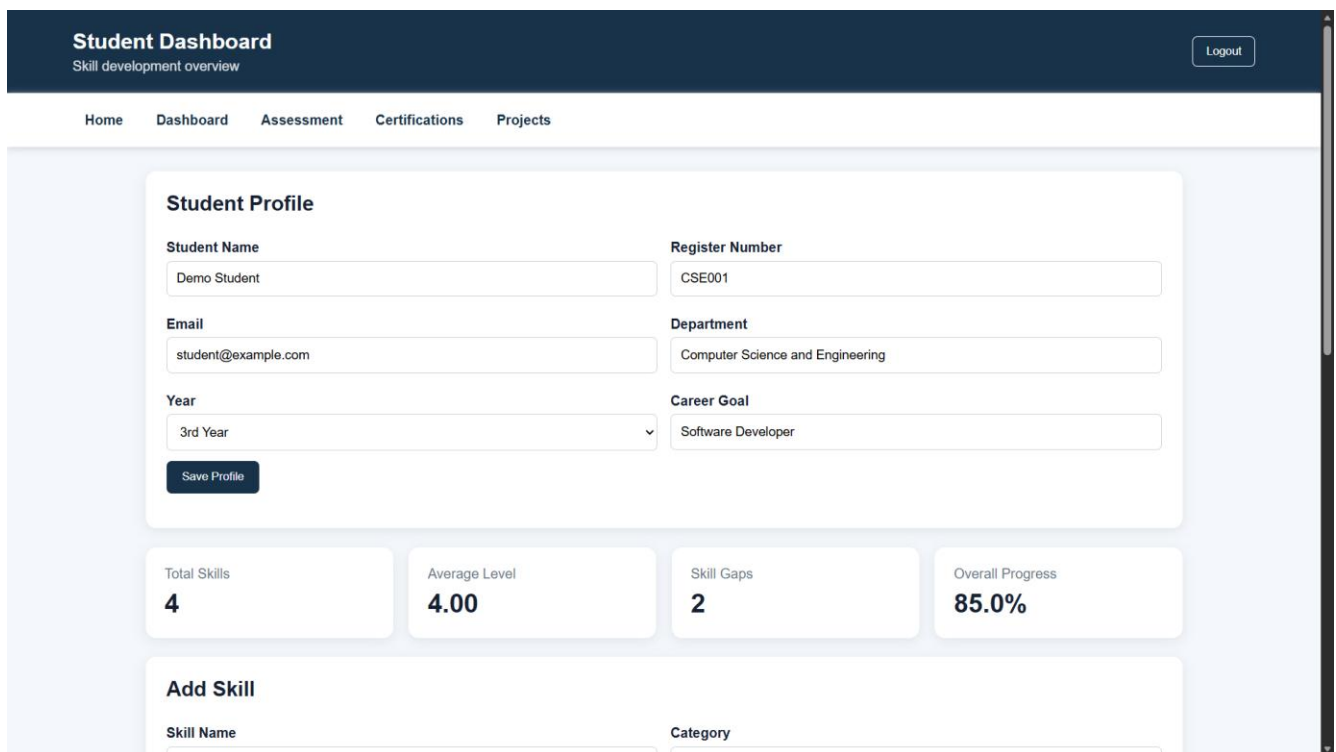
Register Number

[Register](#)

Already registered? [Login](#)

Figure 2 – Registration Page

Output: Student registration form accepts name, email, password and registration number.



The image shows a 'Student Dashboard' with a dark blue header and a light blue background. The header includes the title 'Student Dashboard' and a 'Logout' button. Below the header is a navigation bar with links: 'Home', 'Dashboard', 'Assessment', 'Certifications', and 'Projects'. The main content area features a 'Student Profile' section with a 'Save Profile' button. Below the profile section are four cards displaying 'Total Skills' (4), 'Average Level' (4.00), 'Skill Gaps' (2), and 'Overall Progress' (85.0%). At the bottom is an 'Add Skill' section with input fields for 'Skill Name' and 'Category'.

Student Dashboard
Skill development overview

[Logout](#)

[Home](#) [Dashboard](#) [Assessment](#) [Certifications](#) [Projects](#)

Student Profile

Student Name

Register Number

Email

Department

Year

Career Goal

[Save Profile](#)

Total Skills **4**

Average Level **4.00**

Skill Gaps **2**

Overall Progress **85.0%**

Add Skill

Skill Name

Category

Figure 3 – Dashboard

Output: Dashboard displays total skills, average skill level, skill gaps and overall progress.

Skill Assessment
Evaluate student proficiency

Logout

[Home](#) [Dashboard](#) [Assessment](#) [Certifications](#) [Projects](#)

Assessment Form

Skill

Java

Score (0-100)

Evaluate

Java

Score: 75/100

Level: Intermediate

Assessment History

Date	Skill	Score	Level
1/9/2026	Java	75	Intermediate

CSA4305 – Internet Programming | Assessment

Figure 4 – Skill Assessment

Output: Assessment score is converted into a proficiency level.

Certifications
Record student achievements

Logout

[Home](#) [Dashboard](#) [Assessment](#) [Certifications](#) [Projects](#)

Add Certification

Certificate Name

Organization

Issue Date

dd - mm - yyyy

Certificate ID

Add Certification

Certification Records

Certificate	Organization	Issue Date	ID	Action
Java	Oracle	2026-08-20	1234646	Delete

CSA4305 – Internet Programming | Certifications

Figure 5 – Certification Module

Output: Certification information is displayed in a table.

Projects & Internships
Track practical experience

Logout

Home Dashboard Assessment Certifications Projects

Add Project

Project Name

Technology

Type

Duration

Description

Add Project

Project Records

Project	Technology	Type	Duration	Description	Action
Capstone	HTML	Academic Project			Delete

Figure 6 – Project Module

Output: Project/internship records are displayed.

Year

Career Goal

Save Profile

Total Skills **4**

Average Level **4.00**

Skill Gaps **2**

Overall Progress **85.0%**

Add Skill

Skill Name

Category

Current Level (1-5)

Required Level (1-5)

Add Skill

Skill Analysis

Skill	Category	Current	Required	Gap	Status	Action
Java	Programming	4	5	1	Improvement Required	Delete
Python	Programming	5	5	0	Achieved	Delete

Figure 7 – XML/XSLT

Output: XML skill information is transformed and displayed through XSLT.

10. Results and Validation

The developed prototype successfully performs the major operations required for student skill development management.

Sample Dashboard Result

Metric	Result
Total Skills	4
Average Skill Level	4.00/5
Skills Requiring Improvement	2

Metric	Result
Overall Progress	85%
Assessment Records	1
Certifications	1
Projects	1

Sample Skill Analysis

Skill	Current Level	Required Level	Gap	Status
Java	4	5	1	Improvement Required
Python	5	5	0	Achieved
HTML/CSS	4	4	0	Achieved
JavaScript	3	5	2	Improvement Required

The results demonstrate that the system can identify skills requiring improvement and present the information through the dashboard.

11. Analysis, Comparison, Trade-offs and Justification

Two approaches can be considered for skill-gap analysis.

Approach 1 – Rule-Based Analysis

Skill Gap = Required Level - Current Level

Advantages

- Simple to implement.
- Easy to understand.
- Easy to test.
- Transparent.
- Requires little data.
- Suitable for an academic prototype.

Disadvantage

It does not consider complex patterns in student learning behaviour.

Approach 2 – AI/ML-Based Analysis

An advanced system could use historical assessment and learning data to predict skill requirements and generate personalized recommendations.

Advantages

- More personalized.
- Can analyse larger amounts of data.
- Can identify complex patterns.

Disadvantages

- Requires a suitable dataset.
- More difficult to implement.

- Requires additional computational resources.
- Results can be harder to explain.

Selected Approach

The rule-based approach was selected because the project is an academic prototype and requires a transparent, simple and easily testable methodology.

12. Broader Considerations / SDG Relevance

SDG 4 – Quality Education

The system supports Quality Education by helping students understand their current skill levels and identify areas that require improvement. It encourages continuous learning and structured skill development.

SDG 8 – Decent Work and Economic Growth

The system supports employability by maintaining information about skills, certifications, projects and internships. Students can use this information to identify areas that need improvement before entering the workplace.

SDG 9 – Industry, Innovation and Infrastructure

The project demonstrates the use of digital technologies to create infrastructure for student skill management and development.

The assignment specifically maps the project to SDG 4, SDG 8 and SDG 9.

Privacy and Ethics

Student information should be protected through:

- Authentication
- Authorization
- Input validation
- Controlled access
- Secure storage

The assignment also emphasizes protecting personal information and controlling access to student records and evidence.

13. Conclusion, Limitations and Possible Improvements

Conclusion

The Student Skill Development Management System provides a centralized platform for managing student profiles, skills, assessments, certifications, projects and progress. The system identifies competency gaps by comparing current and required skill levels and provides suitable learning recommendations.

The implementation demonstrates practical use of HTML, CSS, JavaScript, XML and XSLT concepts. The dashboard provides a simple way to understand overall skill-development status.

The prototype therefore provides a useful foundation for a larger institutional student skill-management application.

Limitations

1. The prototype uses LocalStorage instead of a centralized database.
2. A complete JSP/Servlet backend is not implemented.
3. Advanced role-based access control is not implemented.

4. Mentor feedback is not included.
5. Evidence/document upload is not included.
6. Notifications are not implemented.
7. Recommendations use simple rules rather than AI.
8. Advanced graphical analytics are limited.

Possible Improvements

Future versions can include:

- JSP/Servlet MVC architecture.
- MySQL/PostgreSQL database.
- Secure authentication.
- Student/faculty/admin roles.
- Mentor feedback.
- Evidence upload.
- Email notifications.
- AI-based recommendations.
- Skill prediction.
- Advanced analytics.
- Mobile application.
- Cloud deployment.

The assignment also identifies AI-assisted recommendations, competency prediction, mobile access, industry skill frameworks and advanced analytics as possible future directions.

14. Individual Contribution of Group Members

Member 1 – Front-End and User Management

- Analysed the problem statement.
- Designed login and registration pages.
- Developed student profile interface.
- Worked on HTML structure.
- Prepared documentation.

Member 2 – JavaScript and Dashboard

- Developed JavaScript functionality.
- Implemented skill management.
- Implemented skill-gap calculation.
- Developed assessment functionality.
- Developed dashboard and progress calculation.
- Conducted functional testing.

Member 3 – XML/XSLT and Records

- Developed XML skill data.
- Implemented XSLT transformation.

- Developed certification module.
- Developed project/internship module.
- Prepared screenshots and test results.
- Assisted in report preparation.

15. References

1. IP Assignment.pdf – Student Skill Development Management System, Internet Programming course assignment specification.
2. MDN Web Docs – HTML, CSS and JavaScript reference.
3. W3C – HTML and CSS standards.
4. W3C – XML and XSLT specifications.
5. GitHub Documentation – Version control and project collaboration.
6. Google Chrome / Microsoft Edge – Application testing environments.
7. JavaScript LocalStorage – Browser-side data storage.
8. Any external tutorials, libraries, datasets or AI tools actually used during development should be listed separately in the final submission.